Santa Clara University

Mechatronics – ECEN 123/MECH 143/CSEN 123

Homework 1

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Write each of the following sketches in Arduino C. Run them on your Arduino and test them to make sure they produce the correct results. Turn in a printed copy of the source code (PDF) on Camino. Keep the original programs; the TA may ask you to email them in for verification. Include comments in the code.

Review the Arduino Reference website as necessary.

1) Copy the sketch "for_loop" from the Class Examples directory on Google Drive. Run it and observe the output on the serial monitor. (Click the magnifying glass icon in the upper right of the Arduino IDE).

CODE

```
/*
  Hello World
  Prints Hello World to the Serial Port 10 times
 */

void setup() {
  // initialize serial communication at 9600 bits per second:
  int i;
  Serial.begin(9600);
  for (i=1; i<=10; i++) {
    Serial.print("I can count to ");
    Serial.println(i);
  }
}
void loop() {
}
```

RESULTS:

```
Output    Serial Monitor ×
Message (Enter to send mess
I can count to 1
I can count to 2
I can count to 3
I can count to 4
I can count to 5
I can count to 6
I can count to 7
I can count to 8
I can count to 9
I can count to 10
```

Now:

a. Modify this program to use a “while” loop.

CODE

```
/*
  Hello World
  Prints Hello World to the Serial Port 10 times
 */

void setup() {
  // initialize serial communication at 9600 bits per second:
  int i;
  Serial.begin(9600);
```

```
  i=1;
  while(i<=10) {
    Serial.print("I can count to ");
    Serial.println(i);
    i++;
  }
}
void loop() {
}
```

RESULTS:

Output Serial Monitor ×

Message (Enter to send messa

```
I can count to 1
I can count to 2
I can count to 3
I can count to 4
I can count to 5
I can count to 6
I can count to 7
I can count to 8
I can count to 9
I can count to 10
```

b. Modify the program to use a “do while” loop.

CODE

```
/*
  Hello World
  Prints Hello World to the Serial Port 10 times
 */

void setup() {
  // initialize serial communication at 9600 bits per second:
  int i;
  Serial.begin(9600);
  i=1;
   do {
    Serial.print("I can count to ");
    Serial.println(i);
    i++;
  } while(i<=10);
}
void loop() {
}
```

RESULTS:

Output Serial Monitor ×

Message (Enter to send mes

```
I can count to 1
I can count to 2
I can count to 3
I can count to 4
I can count to 5
I can count to 6
I can count to 7
I can count to 8
I can count to 9
I can count to 10
```

2) Write a program that calculates and prints out all of the digits of the first 4 powers of 13 (13^1up to 13^4) separated by commas using a while loop. Make sure your calculated values are *exactly correct*. Include your printed results in the answer.

CODE

```
void setup() {
  // initialize serial communication at 9600 bits per second:
  Serial.begin(9600);
  Serial.println("");
  float i=13.0;
  float count = 1.0; //HW states to start at 13^1
   while (count<5) {
    Serial.print(round(pow(i,count))); //have to round because
arduino floating point precision is buns
    count++;
    if(count < 5) Serial.print(", ");
  }
```

```
}
void loop() {
}
```

RESULTS:

Output Serial Monitor ×

Message (Enter to send messa

```
13, 169, 2197, 28561
```

3) Modify the program so that it correctly calculates and prints out prints out all of the digits of the first 7 powers of 17 separated by commas. Make sure your calculated values are *exactly correct*. Include your printed results in the answer.

Assuming we start from 7^1 like in the previous problem:

CODE

```
void setup() {
  // initialize serial communication at 9600 bits per second:
  Serial.begin(9600);
  Serial.println("");
  long i=17;
  int count = 1; //start at 1 like previous problem
   while (count<8) {
    Serial.print(multiplier(i,count));
    count++;
    if(count < 8) Serial.print(", ");
  }
}
```

```
long multiplier(long input, int count){ //long is necessary for
precision | OBNOXIOUS
  int j = 1;
  long total = input;
  while (j < count) {
    total *= input;
    j++;
  }
  return total;
}
void loop() {
}
```

RESULTS:

Output Serial Monitor ×

Message (Enter to send message to 'Arduino Mega or Mega 2560' on '

```
17, 289, 4913, 83521, 1419857, 24137569, 410338673
```

4) Write a program that repeatedly allows you to type in a string of up to 50 characters on the serial monitor followed by a return. Then print a single line to the serial monitor that says:

```
You entered xx characters.
```

But instead of xx, your output should have the actual count of the characters typed. Spaces and punctuation count. Do not count the enter or return key at the end as a character. Include example printed results in the answer.

CODE

```
void setup() {
  // initialize serial communication at 9600 bits per second:
  Serial.begin(9600);
  Serial.setTimeout(200); //allow person to type
  Serial.println("Type yo message pal (50 char max): ");
}

void loop() {
  if (Serial.available() > 0) {
    String message = Serial.readStringUntil('\n');
    Serial.println("Your message: " + message);
    int x = message.length();
    if (x>50) Serial.println("Too many characters. 50 max!");
    else {
      Serial.println("You entered " + String(x) + " characters.");
    }
    delay(2000); //wait a lil bit to enter new message
    Serial.println("Type yo message pal (50 char max): ");
  }
}
```

RESULTS:

```
Output    Serial Monitor ×
Message (Enter to send message to 'Arduino Mega or Mega 2560' on 'COM3')    New Line
Type yo message pal (50 char max):
Your message: one
You entered 3 characters.
Type yo message pal (50 char max):
Your message: ~!@#$%^&*()_+=-0987654321`qwertyuiop[]\ ';/.,<>:Lj
You entered 50 characters.
Type yo message pal (50 char max):
Your message: ~!@#$%^&*()_+=-0987654321`qwertyuiop[]\ ';/.,<>:Ljc
Too many characters. 50 max!
Type yo message pal (50 char max):
Your message:
You entered 0 characters.
Type yo message pal (50 char max):
Your message: random sentence as last test, surely not ober 50
You entered 48 characters.
Type yo message pal (50 char max):
```

5) Make an improved copy of your last program that also outputs a second line that says:

`yy of those characters were lower-case letters.`

But instead of yy, your output should have the actual count of the lower-case letters typed. Include example printed results in the answer.

CODE

```
void setup() {
  // initialize serial communication at 9600 bits per second:
  Serial.begin(9600);
  Serial.setTimeout(200); //allow person to type
  Serial.println("Type yo message pal (50 char max): ");
```

```
}

void loop() {
  if (Serial.available() > 0) {
    String message = Serial.readStringUntil('\n');
    Serial.println("Your message: " + message);
    int x = message.length();
    int y = 0;
    for(int i = 0; i < x; i++) if(isLowerCase(message[i])) y++;
    if (x>50) Serial.println("Too many characters. 50 max!");
    else {
      Serial.println("You entered " + String(x) + " characters.");
      Serial.println(String(y) + " of those characters were
lower-case letters.");
    }
    delay(2000); //wait a lil bit to enter new message
    Serial.println("Type yo message pal (50 char max): ");
  }
}
```

RESULTS:

Output Serial Monitor ×

Message (Enter to send message to 'Arduino Mega or Mega 2560' on 'COM3')

```
Type yo message pal (50 char max):
Your message: Hello
You entered 5 characters.
4 of those characters were lower-case letters.
Type yo message pal (50 char max):
Your message: HELLO
You entered 5 characters.
0 of those characters were lower-case letters.
Type yo message pal (50 char max):
Your message: hello
You entered 5 characters.
5 of those characters were lower-case letters.
Type yo message pal (50 char max):
Your message: WHAT abOUT ThIS )*$&%#&$I(%":?><}{
You entered 34 characters.
3 of those characters were lower-case letters.
Type yo message pal (50 char max):
Your message: ahfbsjhdbvsjhldbvljhsdfbvlkjdhsbvlhdbsfljvdfjlhbv djfhbvdljfb dlkjfbjld
Too many characters. 50 max!
Type yo message pal (50 char max):
```